Wenlong Ding

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Education

The Chinese University of Hong Kong (CUHK) Shatin, Hong Kong SAR, China
Ph.D. in Computer Science and Engineering *08/2021 – Present*

- Research Topics: ML for network

Huazhong University of Science and Technology (HUST) Wuhan, Hubei, China
B.E. in Computer Science and Technology (ACM class) *09/2017 – 06/2021*

- GPA: 90.64/100 or 3.96/4.0 | Rank: Top 5%

Research Experience & Academic Competition

Energy-efficient In-memory Processing Linear Algebra System 02/2021 – 05/2021
This work has been accepted by Journal of Computer Science and Development (in Chinese) *HUST, China*

- Propose a self-selected mantissa compaction MVM system based on heterogeneous memristive crossbar sets to achieve both high-precision floating point calculation and energy-saving low-precision calculation.
- Design a dynamic alignment bit optimization mechanism that can reduce the extra overhead caused by redundant alignment bits.
- Design a pipelined shift-and-add reduction tree structure with variable number of leaf nodes, which can perform pipelined sum operation while activating any number of mantissa and alignment crossbars.
- Integrate the analog memristive in-situ MVM system into the high-performance linear algebra algorithm framework for evaluation. The evaluation result shows that when the crossbar-based in-memory solutions have average solving residual of $0 \sim 10^{-3}$ order of magnitude compared with the software baseline, the average energy consumption of computational crossbars and peripheral analog-to-digital converters are reduced by $5\% \sim 65\%$ and $30\% \sim 55\%$, respectively.

Disk Failure Prediction Via GAN Anomaly Prediction 09/2020 – 11/2020

- Convert one-dimensional SMART data into 2D SMART data using sliding windows.
- In the training phase, only the 2D SMART data of the normal disk is trained in GAN anomaly prediction module to capture its data characteristics, and the threshold is set according to its Loss.
- In the verification phase, normal disks and abnormal disks are distinguished through the set threshold.
- Achieve a FDR as high as 95% and a FAR as low as 1% in a real world dataset *Backblaze*.

Retention-aware Container Caching in edge computing 04/2019 – 06/2019

- We provide a model to handle C2RD (container caching jointly with request distribution) question, which finds the tradeoff between service latency cost and container retention cost.
- Inspired by the classic ski-rental problem, we present RDC (Random Distribution and Caching) algorithm for multi-node case at one timeslot (a special case).
- We propose O-RDC (Opportunistic-RDC) to handle the general cases (multi-node case at consequent timeslots). It turns out that the model has a relatively good competitive ratio.
- We carry out an experiment on a real-world dataset *EUA*. The model outperforms existing cache strategies by up to 97% in overall system cost.

***The Opioid Crisis*, Problem C of Mathematical Contest in Modeling 2019** 01/2019
This project won the Meritorious Winner awarded by COMAP in 04/2019 *COMAP, USA*

- We should figure out the spreading source and future users distribution of opioid with the given dataset.
- Use SPBFA (Source-Finding Problem Based on Floyd Algorithm) to find the spreading source.

- Determine the edges weights using PCCM (Partial correlation coefficient modified function).
- Use Granger Causality test to extract more possible factors which will influence our model. Combined with PCCM, we recompute the weights in our previous model.
- Use Markov Chain to predict the future trend of opioid.

Other Projects

Mini-C Language compiler with Performance Optimization 05/2020 – 06/2020

- The system translates Mini-C Language (Subset of C Language) to MIPS Assembly language.
- The system can identify all lexical and grammatical errors through lexical and syntax analysis program. Also, the static semantic program can identify 20+ kinds of static semantic errors.
- Construct DAG to achieve local optimization of target code and extract invariant operations to achieve loop optimization. The length of target code has been reduced by 31.3% in the benchmark.
- Implement target code register allocation strategy using DAG heuristic sorting algorithm. The number of registers we use has been reduced by 22.8% in the benchmark.

Performance Optimized Five-stage Pipeline MIPS CPU 03/2020

- Construct controller, pipeline interface components, data path, etc. to form a Pipeline CPU.
- Add data redirection technology and dynamic branch prediction mechanism (BHT prediction module), the runtime of benchmark has been reduced by 36.6% and 49.8% respectively.
- Support two modes of interrupt in our pipeline: hardware interrupt, software interrupt.
- Implement a box pushing game with graphic interface and run on the CPU completed above.

A Web Server Software Supporting Multithreading 12/2019

- Support to configure the listening address, listening port and home directory of the web server.
- Web Server can return the requested page normally, and handle exceptions to error pages.
- Support the transmission of multiple types of files, such as multimedia files.
- Support multi-thread communication and GUI.

Overseas Exchange Experience

Participant of Summer Workshop 07/2019

School of Computing (SoC), National University of Singapore (NUS), Singapore

- Conduct a data processing project about predicting the house price with the dataset in Kaggle.
- Our model gives a good result with RMSE as low as 0.114 and ranked top 4% in Kaggle project.

Acdemic Course About Deep Learning 10/2018

University of California, Los Angeles(UCLA), USA

- Take some courses about the basic knowledge about Deep Learning.
- Conduct a project using CNN to identify the face dataset and the accuracy can reach 98%.

Honors and Awards

06/2021	Excellent Undergraduate Thesis (3%), HUST
05/2021	Outstanding Graduate and Honours Degree (3%), HUST
09/2020	Merit Student (5%), HUST
12/2019	Outstanding Individual in University Students Innovation and Entrepreneurship Project, HUST
04/2019	Meritorious Winner, Mathematical Contest in Modeling (5%), COMAP, USA
03/2018	Freshmen Scholarship (10%), HUST

Technical Skills

Languages: C/C++, Python, html/css/javascript

Software: VScode, Xcode, Microsoft Office, Vivado, Logisim, QT Creator, SQL Server management

Operating System: MacOS, Linux(ubuntu), Windows10